

Mohammadreza Nemati

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EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Optimized Qwen2.5-VL for autonomous-driving trajectory prediction, reducing parameters from 3.75B to 1.4B through depth pruning for deployment under edge compute and memory constraints.
- Distilled knowledge from Qwen2.5-VL 7B into the pruned 1.4B trajectory-prediction model, applying quantization-aware training to preserve VLM performance under constrained deployment settings.
- Built distributed PyTorch/DDP pipelines for multi-task automotive VLM workloads, supporting scalable training, task-specific evaluation, and performance validation across transformer-based models.
- Post-trained multimodal foundation models using SFT, DPO, and GRPO for reasoning, grounding, and task execution in automotive AI workflows.
- Developed multimodal inference and evaluation infrastructure for automotive AI workflows, measuring latency, throughput, task accuracy, and serving reliability across large VLM experiments.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built VLM-based visual grounding system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language object grounding and segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

VLA / Automotive AI: Vision-Language-Action Models, Trajectory Prediction, Autonomous-Driving VLMs, Multimodal Reasoning, Visual Grounding

ML Engineering: Python, PyTorch, Distributed Training (FSDP/DDP), Fine-tuning, Post-training, Reinforcement Learning, Model Evaluation

Optimization / Deployment: Pruning, Knowledge Distillation, Quantization-Aware Training, vLLM, CUDA, Model Serving, Latency/Throughput Tuning

Computer Vision: Vision Transformers, Segmentation, Cross-Modal Alignment, Object Grounding, Graph Neural Networks

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017